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Quantum Machine Learning for Analyzing Astronomical Objects

A fair comparison of quantum and classical models on SDSS spectra

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01

The Problem

Two million spectra

The Sloan Digital Sky Survey (SDSS)

Each SDSS spectrum is a **1D signal**: thousands of flux measurements across wavelength, encoding a source's

- composition & temperature,
- redshift (distance / velocity),
- and its **class**: star, galaxy or quasar.

SDSS has measured **>2 million** such spectra [1] — far too many to classify by hand.

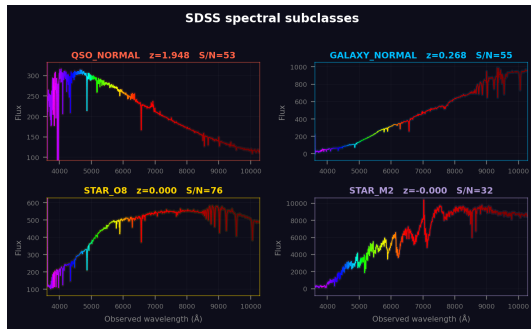


Figure: four spectral subclasses from our curated SDSS DR19 set.

The easy problem is solved

...so we went looking for the hard one

Coarse classification (3 classes)

Star vs. galaxy vs. quasar is **essentially solved**. Modern deep models (e.g. GaSNet-II [2]) reach $\approx 99\%$ accuracy.

Fine-grained subclasses (62 classes)

Accuracy collapses at the boundaries between *physically similar* types — e.g. a late-type M8 dwarf vs. an L brown dwarf.

The driving question

Can **quantum machine learning** add anything useful exactly where strong classical models begin to **struggle**?

A deliberately cautious investigation

What a quantum model is here

A **Variational Quantum Circuit (VQC)** is a circuit whose gate-rotation angles are trained by gradient descent [3] — exploiting an exponentially large state space and entanglement.

- A *purely* quantum model **never trained** on raw spectra (known trainability obstacles [4]).
- So every quantum model we study is **hybrid**: a classical feature stage feeding a small quantum component.
- All circuits run on a **noiseless statevector simulator** [5], ≤ 8 qubits — no claims about real hardware.

We treat an apparent quantum advantage as a **hypothesis to be ruled out**, not a result.

The central idea of this thesis

Fairness as a discipline

Most applied QML places a powerful classical network in front of a tiny circuit — then credits the result to the **quantum** part.

We refuse that shortcut. **Every** quantum model is matched to a classical twin of:

- the same **parameter count**,
- the same **surrounding pipeline**,
- trained under the **identical protocol**.

Our three contributions

1. A competitive 14M-param **classical baseline** (“the Beast”).
2. **Five fair comparisons** across four quantum paradigms.
3. The **comparison itself** — a reusable honesty methodology.

02

Background & Approaches

Variational quantum circuits in one slide

Expressivity vs. trainability

What makes them expressive

Data re-uploading [3]: the input is re-injected at every layer. Even a single qubit can become a universal classifier. A VQC is effectively a **Fourier series** in its input [6].

What makes them hard

Barren plateaus [4]: in deep, wide, random circuits gradients vanish *exponentially* in the qubit count — and training stalls.

Width, depth and trainability trade off against each other — a tension our architecture sweeps probe directly.

Four quantum paradigms — one fair test each

Roadmap of the talk

1 · Trainable circuits (VQC)

A small variational circuit as a classifier **head**.

Exp 1 (binary) · Exp 2 (4-class, frozen features)

2 · Quantum kernels (QSVM)

The circuit as a fixed **feature map** [7, 8] for a support-vector machine.

Exp 3 (few-shot)

3 · Quanvolution

A small circuit **slid across** the signal like a convolutional kernel [9].

Exp 4 (noise robustness)

4 · True QCNN

The whole spectrum embedded into **one quantum state** [10], with coherent pooling.

Exp 5 (fully quantum)

Every paradigm is benchmarked against a parameter-matched classical control.

03

Data & Classical Baseline

From 2 million raw spectra to a clean benchmark

SDSS DR19 curation

- Drop unreliable redshifts ($\text{ZWARNING} > 0$).
- **Merge** > 100 raw subclasses into physically coherent groups (e.g. B3II , $\text{B3V} \rightarrow \text{B3}$).
- Enforce ≥ 250 samples/class; cap at 25 000 to fight imbalance.
- **Withhold** redshift scalars — they let networks cheat; force raw-flux learning.

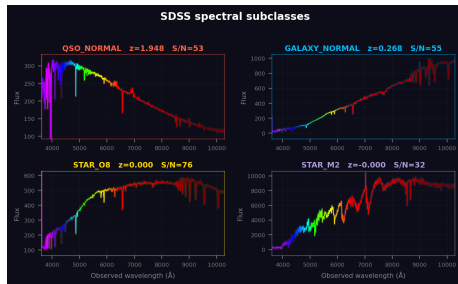


Figure: example classes — QSO, GALAXY, STAR_O8, STAR_M2.

Final dataset

642 588 spectra, 62 classes — a rigorous classical testbed.

The classical baseline: “the Beast”

~14M parameters, built to be hard to beat

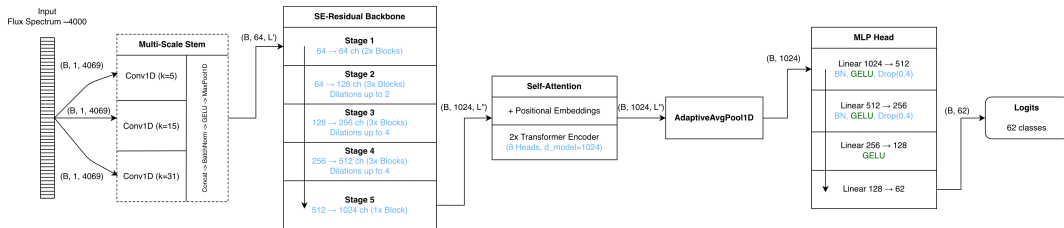


Figure: multi-scale stem $\rightarrow$ dilated SE-residual backbone [11] $\rightarrow$ Transformer encoder [12] $\rightarrow$ MLP head (62 logits).

Why this design

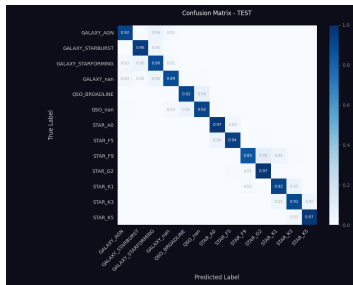
Spectra mix **local** lines and a **global** continuum — so we need both fine extraction and a global receptive field.

Two roles

A competitive baseline *and* the **frozen feature extractor** reused in Experiment 2.

The Beast is competitive with the state of the art

Two validations



13-class GaSNet-II replica [13]: 92.5%.

Headline numbers

- **84.5%** on the full **62-class** problem.
- **92.5%** reproducing the **GaSNet-II** 13-class benchmark.

Training recipe

OneCycle LR [14], 300 epochs, **Focal loss** [15] for imbalance, GELU throughout (no dead neurons).

Letting the baseline pick its own hardest fights

Confusion-driven target selection

An automated analysis of the 62×62 confusion matrix extracts:

- **Mirrors** — symmetric $A \leftrightarrow B$ confusions (e.g. STAR_A5 $\leftrightarrow$ STAR_F0, 33.7% overlap).
- **Black holes** — “sink” classes that absorb false positives (e.g. GALAXY_BROADLINE).

The single hardest pair, STAR_BROWN_DWARF_L vs. STAR_M8, becomes the target for the quantum experiments.

Why this matters

We don't hand the quantum model an easy task. We give it the **exact pairs** a 14M-param network cannot separate.

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Experiment 1 · Binary VQC

Setup: swap only the contested middle

The hardest pair, isolated

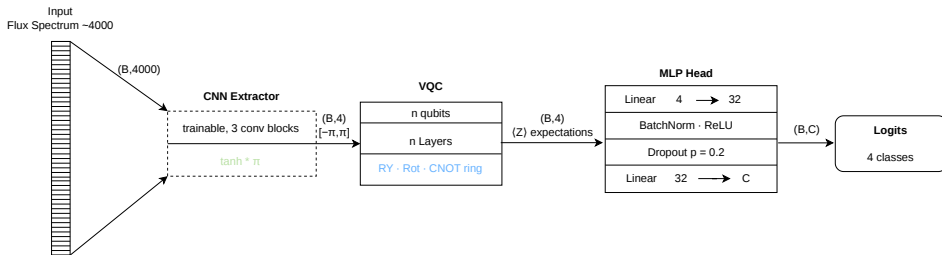


Figure: a trainable CNN extractor compresses the 4096-bin spectrum to a few features in $[-\pi, \pi]$; the middle stage is either a VQC or a parameter-matched classical block; a shared MLP head decides.

The CNN extractor holds $\sim 24\,000$ of $\sim 24\,500$ parameters. The contested circuit is just **72 angles** — so any difference is localised to that one stage.

Result: parity — and a class-balance surprise

L brown dwarf vs. M8

Model	Config	Acc.	BROWN rec	Gap
Quantum	2q×12L	0.863	0.70	0.232
Quantum	4q×6L	0.869	0.73	0.200
Quantum	8q×3L	0.860	0.80	0.083
Classical	4 feat.	0.868	0.77	0.136

Mean over 3 seeds; full sweep $\pm$ std in backup. Gap = recall difference between the two classes.

Top-line

Best quantum (0.869) and best classical (0.868) are within **one std** — no advantage.

The one quantum-specific finding

Wide-shallow circuits balance the classes best (gap 23% $\rightarrow$ 8%) — against the barren-plateau expectation.

Lesson #1: the advantage that wasn't

How an apparent quantum win dissolved

The early result

A first run showed a *large* quantum lead — the classical mirror used a **ReLU** block that collapsed onto the majority class.

After a fair rebuild

Swapping the mirror's activation to tanh closed the gap entirely. The two models tied.

Takeaway

The “advantage” lived in an **under-built baseline**, not in the quantum circuit.

Spiky validation curves looked like a barren plateau — but they appear *just as strongly* in the classical mirror, so they are model-selection noise, not a quantum pathology.

05

Experiment 2 · Frozen-Feature Heads

Setup: four heads, one frozen Beast

Isolating the head, four classes

The frozen Beast turns every spectrum into the **same** 128-dim vector. Only the small head on top changes:

- **A** — dense classical, **5058** params.
- **B** — hybrid **quantum** (Linear+VQC), 556.
- **C** — classical mirror, **ReLU**, 556.
- **D** — classical mirror, tanh, 556.

B, C, D share the *exact* shape & parameter count — isolating the circuit, not the structure.

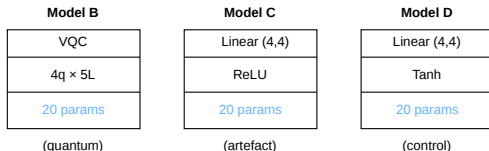
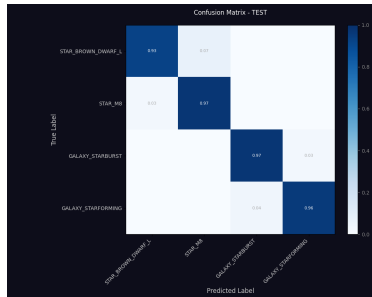


Figure: the three interchangeable 20-parameter middle stages (B quantum, C ReLU, D tanh) in an otherwise identical pipeline.

Result: quantum ties the dense head at $9\times$ fewer params ...and one control collapses

	Head	Params	Acc.
A	Dense	5058	$\sim 96\%$
B	Quantum	556	$\sim 96\%$
C	Cl. ReLU	556	$\sim 79\%$
D	Cl. tanh	556	$\sim 96\%$

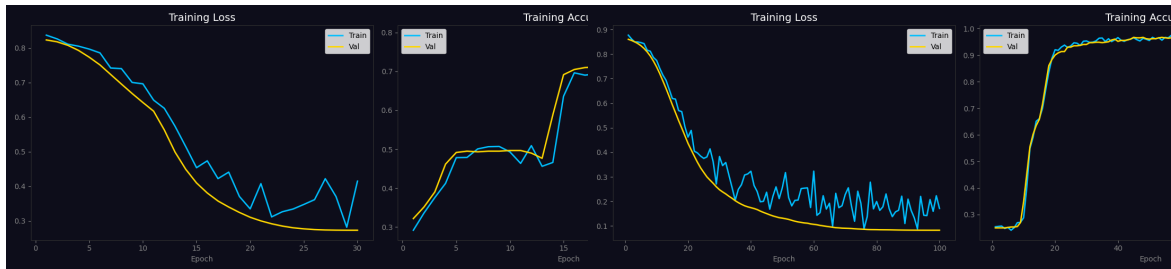
A, B and D are nearly indistinguishable; only the ReLU control fails.



Quantum head (B): the hardest class, STAR_BROWN_DWARF_L, recalled at 0.93.

Lesson #2: a 17-point swing from one activation

The dead-ReLU artefact



ReLU (C): stuck near 50% for many epochs → 79%.

tanh (D): clean convergence → 96%.

With only 4 neurons, a ReLU pushed negative **dies** (zero output, zero gradient). The single largest accuracy swing of the experiment came from the **activation function** — not from quantum vs. classical.

Where the real saving comes from

Encoding > qubit count; extractor > head

Encoding ablation (same 4 qubits)

Encoding into VQC	Params	Acc.
Frozen PCA(128,4)	~40	~76%
Frozen PCA + Lin.	~300	~76%
Learned Linear	556	~ 96%

Which four numbers enter the circuit matters — not the four-qubit count.

The honest reading

The $9\times$ parameter saving is **real** — but it belongs to the strong frozen extractor, since the classical tanh head gets the *same* saving.

Cost

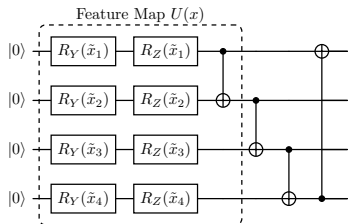
The quantum head is **seconds/epoch** (statevector simulation) vs. **milliseconds** for the classical heads.

06

Experiment 3 · Quantum Kernels

Setup: the circuit as a fixed feature map

Do quantum kernels win when data is scarce?



Feature map $U(x)$: R_Y/R_Z rotations + a CNOT ring.

Bounded encoding $\tilde{x} = \frac{\pi}{2} \tanh(cx)$ avoids 2π wrap-around and kernel concentration.

Two quantum kernels

Fidelity: $|\langle 0|U^\dagger(x')U(x)|0\rangle|^2 = \mathcal{O}(N^2)$ circuits.

Projected (PQK): local Paulis $\rightarrow$ RBF $= \mathcal{O}(N)$.

A pre-registered, honest protocol

Balanced subsets · 95% CIs over seeds · an RBF baseline tuned just as hard · advantage declared *only* if the quantum CI strictly clears the classical CI.

Result: no advantage — and we knew in advance

Classical RBF matches or wins

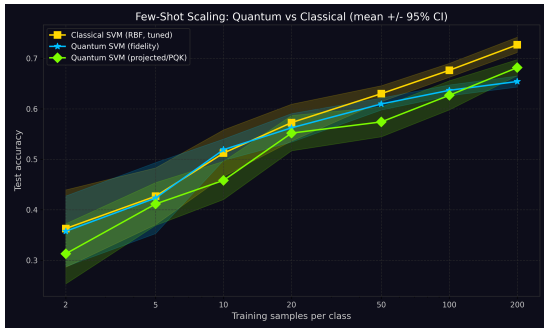


Figure: accuracy vs. training samples per class; mean $\pm$ 95% CI.

- No significant gap for $n \leq 20$.
- Classical RBF **wins** for $n \geq 50$ ($p < 0.05$).

Predicted before training

Geometric difference [16]
 $g(C\|Q) = 1.89$ (fidelity) — near the regime where the classical kernel already reproduces the quantum geometry.

07

Experiment 4 · Quanvolution

Setup: a quantum filter slid across the signal

Does unitarity buy noise robustness?

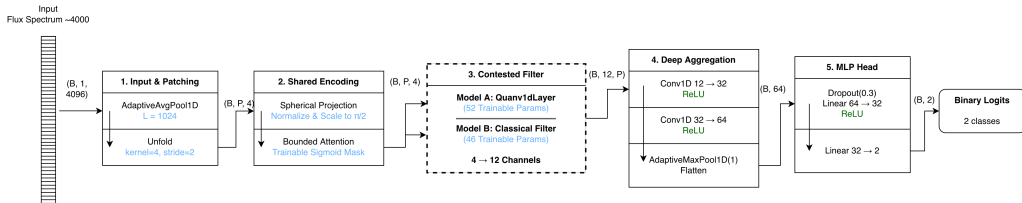


Figure: pool → patch → spherical encoding + bounded attention → contested filter (quantum or classical) → shared deep head.

The hypothesis

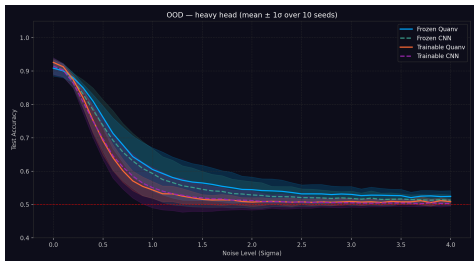
Unitary dynamics $\Rightarrow$ a restricted Lipschitz constant [17] $\Rightarrow$ intrinsic resistance to perturbations.

Strict parity

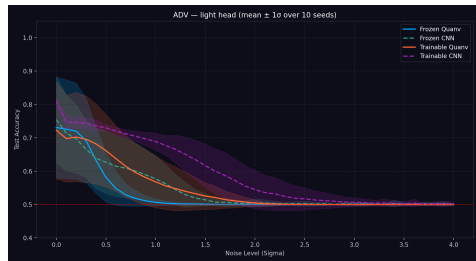
52 quantum params vs. **46** classical.
Tested under OOD & adversarial noise, with a heavy and a light head.

Result: indistinguishable — and the one gap favours classical

Robustness decay, 10 seeds



Heavy head: curves overlap ($\Delta = 0.000$, $p = 0.625$).



Light head, adversarial: classical ahead ($\Delta = -0.047$, $p = 0.014$).

A **frozen random** circuit does as well as the trained one — the 52 variational params add little beyond a fixed embedding. The deep classical head does the work.

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Experiment 5 · True QCNN

Setup: a fully coherent quantum network

No classical extractor in front

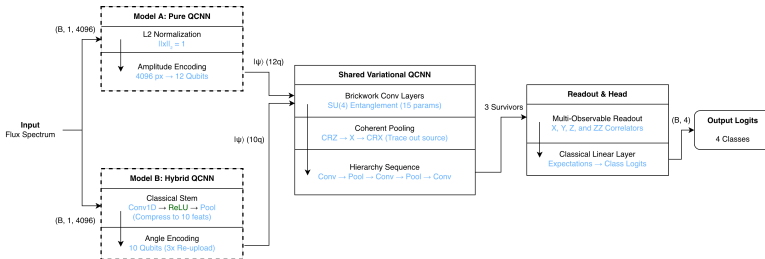


Figure: amplitude-encode 4096 bins into 12 qubits $\rightarrow$ $SU(4)$ convolution + coherent pooling $\rightarrow$ 3 qubits. ~ 101 params.

Designed for quantum data

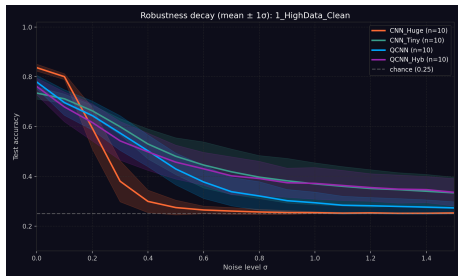
Barren-plateau free [18] — but built for quantum states, not classical flux.

Controls + ablation

Tiny CNN (~ 100 p, strict parity), **Huge CNN** (8228 p, upper bound), and a **Hybrid**

Result: where the quantum inductive bias actually pays off

Robustness across four quadrants



Clean, high-data: the Huge CNN starts highest but **decays fastest** — it overfits.

No *absolute* advantage

The unconstrained Huge CNN wins once scaling and adversarial training are allowed.

But under a strict budget

At ~100 params the QCNN **crushes** the matched Tiny CNN and is the most reliable **few-shot** learner [19].

Fragility = amplitude encoding; the angle-encoded Hybrid QCNN is uniformly more robust.

09

Conclusions & Outlook

The scorecard

Five fair comparisons, one verdict

Paradigm	Task	Verdict vs. matched classical
Exp 1 · Binary VQC	hardest pair	Parity (early lead was a weak baseline)
Exp 2 · Quantum head	4-class, frozen feats.	Parity (saving belongs to extractor)
Exp 3 · QSVM kernels	few-shot	No advantage (classical wins as $n \uparrow$)
Exp 4 · Quanvolution	noise robustness	No advantage (one sig. diff: classical)
Exp 5 · True QCNN	fully quantum	Parity ; edge only at strict parity / few-shot

Quantum models **matched** a carefully built classical model — and at the scales reachable on near-term devices, never beat it.

Parity, not advantage.

And because every circuit was simulated without noise, this is the *best case* for quantum — so the null result is conservative.

Why? The hard work is classical

Interpretability confirms it

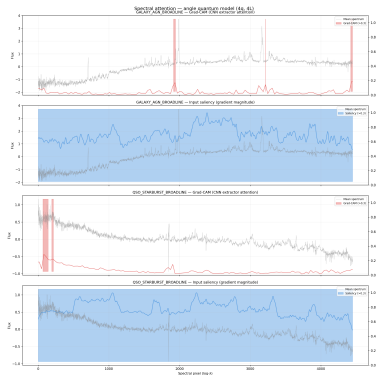


Figure: Grad-CAM localises to broad emission lines and continuum slope — the features an astronomer uses.

The quantum circuit only sees **4 compressed numbers** — never the wavelength axis. The Bayes-optimal discriminant $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$ is a function of the *raw* spectrum, so only the **classical front end** can represent it.

Find the features (classical); then any small head — quantum or classical — can decide.

The contribution is the comparison

A reusable honesty discipline

What we deliver

- Parameter matching at the contested stage.
- Shared pipelines & identical protocols.
- **Robust** classical controls (no dead-ReLU traps).
- Multiple seeds, CIs, Wilcoxon tests.
- Cheap **pre-screening** ($g(C\|Q)$).

Twice, an “advantage” dissolved

Once it was an **under-built baseline**; once a **dead-ReLU artefact**. A single fix erased each.

Treat an apparent quantum advantage as a **hypothesis to rule out** — the classical control matters more than the circuit.

Outlook: where a real advantage might hide

Next steps

- **Change the problem, not the circuit.** Optical spectra are smooth signals classical nets handle naturally. A fair test may need data that is *provably hard* classically — or **quantum in origin**.
- **Confront real hardware.** Repeat on physical devices with error mitigation; check whether structural traits (e.g. smoother feature maps) survive noise.
- **Scale the qubits** and apply the same framework to other signals, to see whether this parity is specific to SDSS spectra.
- **Pre-screen first.** The geometric-difference metric is a cheap, principled filter for deciding which problems are even worth the effort.

None of these is likely to overturn the finding on today's hardware — but each sharpens *where*, if anywhere, a near-term advantage could live.

10

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Thank you!

No quantum advantage — but a method to prove it.

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Questions?

11

Backup Slides

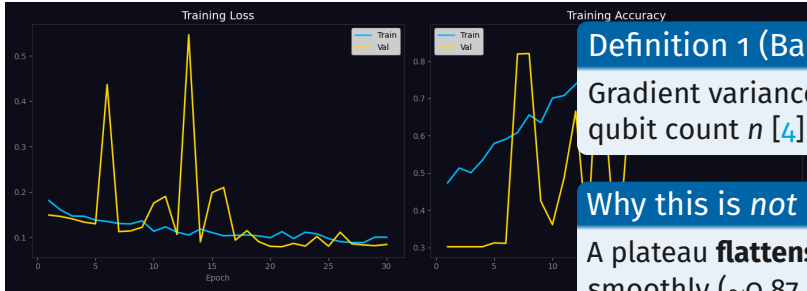
Backup index

What's parked here for your questions

1. Barren plateaus — did we hit one?
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4. Parameter matching: angle $\neq$ weight
5. Fisher discriminant: why classical wins
6. Geometric difference $g(C\|Q)$
7. Fidelity vs. projected quantum kernels
8. Amplitude vs. angle encoding
9. Quantvolution circuit & Jacobian probe
10. QCNN: barren-plateau immunity & few-shot
11. Compute cost & the noiseless caveat
12. The 62×62 confusion analysis
13. Data choices: leakage, imbalance, augmentation
14. Full result tables (Exp 1, 4, 5)
15. Jacobian smoothness slopegraphs

1 · Barren plateaus — did we hit one?

Diagnosing the spiky validation curves



Exp 1 quantum model: training smooth, validation spiky.

Definition 1 (Barren plateau)

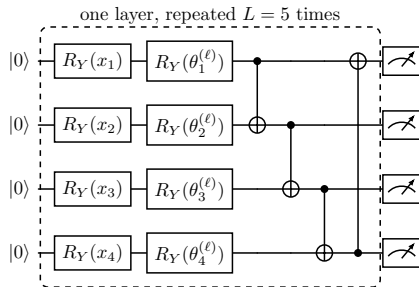
Gradient variance vanishes $\propto 2^{-n}$ in the qubit count n [4] — training stalls.

Why this is *not* one

A plateau **flattens training**. Here loss falls smoothly (~ 0.87 acc.) and the spikes appear *equally* in the classical mirror — so it is model-selection noise, not a quantum effect.

2 · Data re-uploading & the Fourier view

Why a shallow circuit can be expressive



Each layer re-uploads x via $R_Y(x_q)$, then trainable $R_Y(\theta) +$ a CNOT ring; $L = 5$.

Re-uploading [3]

Re-injecting the input at every layer lets a shallow circuit reach higher-order functions; even one qubit becomes a universal classifier.

Fourier picture [6]

A VQC output is a **Fourier series** in x ; the encoding gates fix the accessible frequencies. So the encoding *bounds* what the model can represent — which is why our encoding ablation mattered.

3 · The dead-ReLU mechanism

How a 556-param control fell to 79%

With a width-4 middle layer, a ReLU unit driven negative outputs 0 and receives 0 gradient:

$$\text{ReLU}(z) = \max(0, z), \quad \frac{\partial}{\partial z} \text{ReLU}(z) = 0 \text{ for } z < 0.$$

It **never recovers**. Losing even 1–2 of 4 units destroys the minority-class boundary. $\tanh$ always has nonzero gradient, so it trains cleanly.

Consequence

75% of brown dwarfs were routed into GALAXY_STARFORMING; recall fell to 0.18.

Never report a VQC result without a parameter-matched classical control that uses a **robust** activation.

4 · Parameter matching: an angle is not a weight

Why we matched by construction

The objection

A trainable rotation angle and a classical weight are *not* equivalent units of capacity — so “same parameter count” could be meaningless.

Our answer: structural mirrors

We do not equate units. We hold the **entire pipeline fixed** and swap *only* the contested middle block for one with the same parameter count and the same I/O shape (see Exp 2 models B/C/D). Any measured difference is then localised to that block — sidestepping the angle-vs-weight problem entirely.

We also report **beyond** parameters: convergence speed, stability, per-class recall, and wall-clock cost.

5 · Fisher discriminant: why the front end does the work

A formal “why”

Model a preprocessed spectrum $x \in \mathbb{R}^L$ with Gaussian class-conditionals $x | y=c \sim \mathcal{N}(\mu_c, \Sigma)$. The Bayes-optimal rule is **linear**:

$$w^* = \Sigma^{-1}(\mu_1 - \mu_0), \quad w_i^* = \frac{\mu_{1,i} - \mu_{0,i}}{\sigma_i^2}, \quad J_i = \frac{(\mu_{1,i} - \mu_{0,i})^2}{\sigma_i^2}.$$

The per-bin Fisher ratio J_i is maximised exactly at **broad emission lines** and the **continuum slope** — the regions Grad-CAM highlights.

w^* is a function of the *raw* spectrum. Only the classical extractor sees the wavelength axis, so only it can learn w^* — the head (quantum or classical) merely uses the result.

6 • Geometric difference $g(C\|Q)$

A pre-training advantage filter

$$g = \|\sqrt{K_Q} (K_C + \lambda I)^{-1} \sqrt{K_Q}\|_2$$

Computed on Gram matrices *before* fitting [16].

- $g \approx 1$: the classical kernel reproduces the quantum geometry $\Rightarrow$ **no advantage possible**.
- Large g is **necessary but not sufficient**.

Our values

- Fidelity kernel: $g = 1.89$ — near the no-advantage regime.
- Projected kernel: $g = 12.01$ — distinct, yet *still* no empirical gain.

Both consistent with the measured few-shot parity.

7 · Fidelity vs. projected quantum kernels

Two ways to read a circuit as a kernel

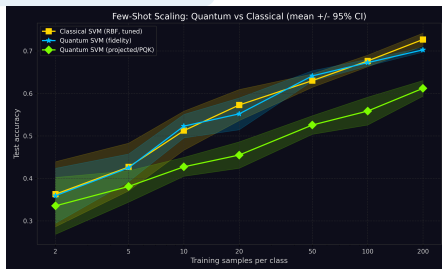
Fidelity kernel

$$k_F(x, x') = |\langle 0 | U^\dagger(x') U(x) | 0 \rangle|^2$$

Geometrically intuitive but $O(N^2)$ circuit evaluations.

Projected kernel (PQK)

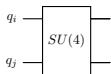
Measure local Pauli X, Y, Z per qubit $\rightarrow$ a $3N$ -dim classical vector, then an RBF on top. Only $O(N)$ circuits.



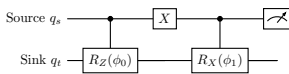
8 • Amplitude vs. angle encoding

The QCNN's real weak point

(a) Convolution Block $U_c(\theta)$



(b) Pooling Block $V_p(\phi)$



QCNN blocks: $SU(4)$ convolution + coherent pooling
(trace out source qubit).

Amplitude encoding

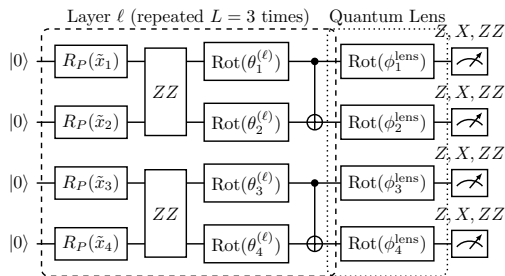
The raw 4096-bin signal is loaded as a global L_2 -normalised state.
High-frequency noise reshapes the *whole* vector multiplicatively — fragile.

Angle encoding (Hybrid QCNN)

A small classical stem $\rightarrow$ 10 bounded features $\rightarrow$ localised rotations. Same conv/pool blocks, **uniformly more robust** in all four quadrants.

9 · Quanvolution circuit & the Jacobian probe

Quantum Lens + smoothness



Multi-axis re-uploading, IsingZZ ring, then a trainable **Quantum Lens**; read out via $Z, X, Z \otimes Z$ (12 channels).

We probe local Lipschitz smoothness via the Jacobian $J_{ij} = \partial f_i / \partial x_j$, reporting median $\|J\|_F$.

Finding

Heavy head: quantum filter *is* smoother ($p = 0.006$). But the ordering **reverses** under the light head — so smoothness is a property of the **learned weights**, not the circuit topology. It buys no accuracy.

10 · True QCNN: immunity & sample efficiency

When the inductive bias pays off

Barren-plateau immunity

True QCNNs are provably free of barren plateaus [18] — but that guarantee was designed for **quantum data**. Embedding classical flux removes the premise, so no absolute advantage appears.

Generalization under a budget [19]

Generalization error scales with **gate count**. The 101-gate QCNN keeps working at 20 samples/class, while the matched Tiny CNN overfits and collapses to chance. The quantum bias manifests as **sample efficiency**, not scalable representational power.

11 · Compute cost & the noiseless caveat

Reading the null result correctly

Cost

Each forward/backward pass is a **statevector simulation**: seconds/epoch on CPU vs. milliseconds for classical heads. Real hardware adds shot noise and decoherence on top.

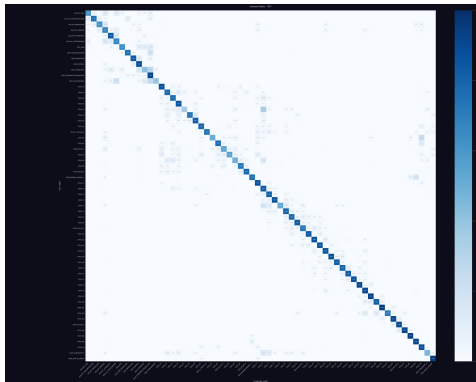
Why noiseless strengthens the result

Simulation is the **best possible case** for the quantum side. Real devices can only *reduce* performance — so finding no advantage here is a **conservative** conclusion.

Scope: ≤ 8 qubits, one scientific domain (SDSS), a handful of hard pairs. We test one specific hybrid regime — not QML in general.

12 • The 62×62 confusion analysis

Mirrors and black holes



Row-normalised 62-class confusion matrix.

Automated failure mining

- **Mirrors:** $A5 \leftrightarrow F0$ (33.7%),
STARBURST $\leftrightarrow$ SB_BROADLINE.
- **Black holes:** GALAXY_BROADLINE
(35% precision, 306 false positives).

This is what *selected* the quantum experiments' target classes — objectively, not by hand.

13 · Data choices: leakage, imbalance, augmentation

Defensible by design

No redshift leakage

Auxiliary scalars (redshift) let networks *trivially* separate classes. We withhold them — models learn from flux alone.

Imbalance

Weighted sampling was unstable; **Focal loss** down-weights easy majority classes and stabilised training.

On-the-fly augmentation

- Class-aware redshift (circular) shift
- Gaussian read-noise + flux scaling
- Contiguous masking (spectral dropout)
- Gaussian smoothing (seeing)
- Mixup ($\lambda \sim \text{Beta}(0.4, 0.4)$)

14a · Full sweep — Experiment 1

Mean $\pm$ std over 3 seeds

Model	Config	Accuracy	Macro-F1	BROWN rec	ROC-AUC
Quantum	2q×12L	0.863 $\pm$ 0.005	0.830 $\pm$ 0.005	0.70	0.910
Quantum	3q×8L	0.865 $\pm$ 0.012	0.835 $\pm$ 0.009	0.72	0.913
Quantum	4q×6L	0.869 $\pm$ 0.008	0.840 $\pm$ 0.005	0.73	0.927
Quantum	6q×4L	0.860 $\pm$ 0.012	0.833 $\pm$ 0.013	0.76	0.924
Quantum	8q×3L	0.860 $\pm$ 0.013	0.837 $\pm$ 0.014	0.80	0.927
Classical	4 features	0.868 $\pm$ 0.016	0.843 $\pm$ 0.016	0.77	0.933
Classical	8 features	0.862 $\pm$ 0.004	0.836 $\pm$ 0.006	0.77	0.922

All configs $\approx$ 24500 params, dominated by the shared CNN extractor. Wide-shallow (8q×3L) is the most class-balanced.

14b · Full results — Experiments 4 & 5

Robustness AUC, 10 seeds

Exp 4 · Quanvolution

Regime	Head	AUC Q	AUC C
OOD	Heavy	0.563	0.563
OOD	Light	0.549	0.564
ADV	Heavy	0.655	0.675
ADV	Light	0.545	0.592

Only Adv-Light significant ($p = 0.014$, favours C).

Exp 5 · True QCNN (best AUC/quadrant)

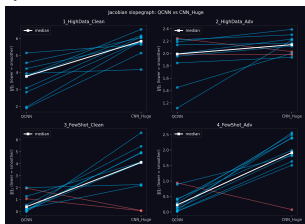
Model	Q1	Q2	Q3	Q4
CNN Huge	0.346	0.673	0.305	0.479
CNN Tiny	0.462	0.544	0.283	0.286
QCNN	0.408	0.576	0.312	0.364
QCNN Hyb	0.448	0.632	0.362	0.443

Q1 clean / Q2 adv (high-data); Q3/Q4 few-shot. Huge wins when scaling is allowed; QCNNs win at parity.

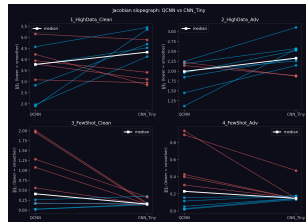
15 · Jacobian smoothness slopegraphs

A second robustness probe (Exp 4 & 5)

Median Jacobian norm of the input→logit map — smaller = smoother.



QCNN (3.52) smoother than the over-parameterised Huge CNN (5.73), which decays fastest.



Few-shot: Tiny CNN's near-zero norm (~ 0.18) is *collapse*, not robustness.

Regime-dependent (Exp 4's ordering even reversed between heads) — it reflects the *learned weights*, not the topology, and buys no advantage.